

GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

Section 3.1 — Deploying and Implementing Compute Engine Resources

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Study Notes Study Notes April 2026

Exam Relevance

This topic is part of **Section 3: Deploying and implementing a cloud solution (~25 % of the exam)**. You must know how to launch compute instances, configure disks and availability policies, manage SSH keys, create autoscaled managed instance groups, configure OS Login, and configure VM Manager.

1. Launching a Compute Instance

Basic VM Creation

```
# Minimal VM creation
gcloud compute instances create my-vm \
  --zone=us-central1-a \
  --machine-type=e2-medium \
  --image-family=debian-12 \
  --image-project=debian-cloud

# Full-featured VM creation
gcloud compute instances create my-vm \
  --zone=us-central1-a \
  --machine-type=n2-standard-4 \
  --image-family=debian-12 \
  --image-project=debian-cloud \
  --boot-disk-size=50GB \
  --boot-disk-type=pd-ssd \
  --tags=http-server,https-server \
  --metadata=startup-script='#!/bin/bash
  apt-get update
  apt-get install -y nginx' \
  --service-account=my-sa@project.iam.gserviceaccount.com \
  --scopes=cloud-platform \
  --labels=environment=production,team=backend \
  --network=my-vpc \
  --subnet=my-subnet \
  --no-address # No external IP
```

Key Creation Parameters

Parameter	Description	Example
<code>--zone</code>	Zone to create the VM in	<code>us-central1-a</code>
<code>--machine-type</code>	CPU/memory configuration	<code>e2-standard-4</code> , <code>custom-6-20480</code>
<code>--image-family</code>	OS image family	<code>debian-12</code> , <code>ubuntu-2204-lts</code>
<code>--image-project</code>	Project containing the image	<code>debian-cloud</code> , <code>ubuntu-os-cloud</code>
<code>--boot-disk-size</code>	Boot disk size in GB	<code>50GB</code>
<code>--boot-disk-type</code>	Disk type	<code>pd-standard</code> , <code>pd-balanced</code> , <code>pd-ssd</code>
<code>--tags</code>	Network tags for firewall rules	<code>http-server,https-server</code>
<code>--metadata</code>	Key-value metadata	<code>startup-script=...</code>
<code>--service-account</code>	Service account for the VM	<code>sa@project.iam.gserviceaccount.com</code>
<code>--scopes</code>	API scopes (legacy, prefer IAM)	<code>cloud-platform</code>
<code>--labels</code>	Labels for organization	<code>env=prod,team=web</code>
<code>--network</code>	VPC network	<code>my-vpc</code>
<code>--subnet</code>	Subnet	<code>my-subnet</code>
<code>--no-address</code>	No external IP	(flag)

2. Assigning Disks

Boot Disk

- Every VM has exactly **one boot disk**
- Contains the operating system
- Defaults: 10 GB for Linux, 50 GB for Windows
- Can specify size, type, and image at creation

Additional Disks

You can attach additional persistent disks to a VM for data storage.

```
# Create a standalone persistent disk
gcloud compute disks create my-data-disk \
  --size=200GB \
  --type=pd-ssd \
  --zone=us-central1-a

# Attach the disk to an existing VM
gcloud compute instances attach-disk my-vm \
  --disk=my-data-disk \
  --zone=us-central1-a

# Create a VM with an additional disk at creation time
gcloud compute instances create my-vm \
  --zone=us-central1-a \
  --machine-type=e2-standard-4 \
  --create-disk=name=data-disk,size=200GB,type=pd-ssd,auto-delete=no
```

Disk Operations Inside the VM

```
# After attaching, format and mount the disk inside the VM
# List attached disks
sudo lsblk

# Format the disk (first time only)
sudo mkfs.ext4 -m 0 -E lazy_itable_init=0,lazy_journal_init=0 /dev/sdb

# Create mount point and mount
sudo mkdir -p /mnt/data
sudo mount /dev/sdb /mnt/data

# Add to fstab for persistence across reboots
echo '/dev/sdb /mnt/data ext4 defaults 0 2' | sudo tee -a /etc/fstab
```

Disk Type Selection

Type	IOPS	Throughput	Cost	Use Case
<code>pd-standard</code>	0.75 R / 1.5 W per GB	Low	Lowest	Logs, bulk storage
<code>pd-balanced</code>	6 R / 6 W per GB	Medium	Medium	General purpose (default)
<code>pd-ssd</code>	30 R / 30 W per GB	High	Higher	Databases, random I/O
<code>pd-extreme</code>	Up to 120K R / 120K W	Highest	Highest	Mission-critical DBs

Key Disk Concepts

- **Persistent disks are network-attached** — survive VM deletion (if auto-delete is off)
- **Local SSDs are physically attached** — data lost on VM stop/termination
- **Auto-delete:** By default, boot disk is auto-deleted with VM; additional disks are not
- **Read-only mode:** A persistent disk can be attached to multiple VMs in read-only mode
- **Resize:** You can increase disk size (never decrease) without downtime

3. Availability Policy

Automatic Restart

- VMs can be configured to **automatically restart** after maintenance events or crashes
- Default: **ON** (recommended for production)
- Set `--restart-on-failure` flag

Host Maintenance Behavior

When Google needs to perform maintenance on the physical host:

Policy	Behavior	Use Case
MIGRATE (default)	Live-migrates the VM to another host	Production VMs, no downtime
TERMINATE	Stops the VM during maintenance	Spot VMs, GPUs (cannot live-migrate)

```
# Set availability policy
gcloud compute instances create my-vm \
  --zone=us-central1-a \
  --machine-type=e2-standard-4 \
  --maintenance-policy=MIGRATE \
  --restart-on-failure

# VMs with GPUs must use TERMINATE
gcloud compute instances create gpu-vm \
  --zone=us-central1-a \
  --machine-type=n1-standard-4 \
  --accelerator=type=nvidia-tesla-t4,count=1 \
  --maintenance-policy=TERMINATE
```

Sole-Tenant Nodes

- Dedicated physical servers for your VMs only
- Use cases: Licensing compliance (BYOL), security isolation, performance
- VMs on sole-tenant nodes still follow maintenance policies

4. SSH Keys

Methods to Connect to VMs

1. OS Login (Recommended)

- Uses IAM to manage SSH access
- Links Linux user accounts to Google identities
- Supports 2FA (two-factor authentication)
- No need to manage individual SSH keys

```
# Enable OS Login on a project
gcloud compute project-info add-metadata \
  --metadata enable-oslogin=TRUE

# Enable OS Login on a specific instance
gcloud compute instances add-metadata my-vm \
  --metadata enable-oslogin=TRUE

# Connect (IAM role required: roles/compute.osLogin or roles/compute.osAdminLogin)
gcloud compute ssh my-vm --zone=us-central1-a
```

2. Metadata-Managed SSH Keys

- SSH public keys stored in project or instance metadata
- **Project-wide keys** — Apply to all VMs in the project
- **Instance-level keys** — Apply to a specific VM only

```
# Add a project-wide SSH key
gcloud compute project-info add-metadata \
  --metadata-from-file ssh-keys=public_keys_file

# Add an instance-level SSH key
gcloud compute instances add-metadata my-vm \
  --metadata-from-file ssh-keys=public_keys_file

# Block project-wide keys on a specific instance
gcloud compute instances add-metadata my-vm \
  --metadata block-project-ssh-keys=TRUE
```

3. gcloud compute ssh

- Simplest method — handles key generation and transfer automatically
- Generates a key pair if one doesn't exist
- Stores the public key in project metadata

```
# Connect to a VM
gcloud compute ssh my-vm --zone=us-central1-a

# Connect with a specific user
gcloud compute ssh alice@my-vm --zone=us-central1-a

# Connect through IAP tunnel (VM has no external IP)
gcloud compute ssh my-vm --zone=us-central1-a --tunnel-through-iap
```

4. SSH via Identity-Aware Proxy (IAP)

- Tunnel SSH connections through Google's IAP service
- VM does **not need an external IP**
- Requires IAM permission: `roles/iap.tunnelResourceAccessor`
- Firewall rule must allow TCP from `35.235.240.0/20` on port 22

5. Managed Instance Groups (MIGs)

What Is a Managed Instance Group?

- A group of **identical VM instances** managed as a single entity
- Based on an **instance template**
- Provides **autoscaling**, **autohealing**, **rolling updates**, and **load balancing**

Instance Templates

An instance template is a **read-only blueprint** for creating VMs:

```
# Create an instance template
gcloud compute instance-templates create my-template \
  --machine-type=e2-standard-4 \
  --image-family=debian-12 \
  --image-project=debian-cloud \
  --boot-disk-size=50GB \
  --boot-disk-type=pd-balanced \
  --tags=http-server \
  --metadata=startup-script='#!/bin/bash
  apt-get update && apt-get install -y nginx'

# List instance templates
gcloud compute instance-templates list

# Describe a template
gcloud compute instance-templates describe my-template
```

Key properties of instance templates:

- **Immutable** — Cannot be edited after creation (create a new one instead)
- Define machine type, image, disks, network, metadata, labels, service account
- Can reference **custom images** or **public images**

Creating a MIG

```
# Create a managed instance group
gcloud compute instance-groups managed create my-mig \
  --template=my-template \
  --size=3 \
  --zone=us-central1-a

# Create a regional MIG (spans multiple zones)
gcloud compute instance-groups managed create my-regional-mig \
  --template=my-template \
  --size=6 \
  --region=us-central1 \
  --target-distribution-shape=EVEN
```

Autoscaling

```
# Configure autoscaling based on CPU utilization
gcloud compute instance-groups managed set-autoscaling my-mig \
  --zone=us-central1-a \
  --min-num-replicas=2 \
  --max-num-replicas=10 \
  --target-cpu-utilization=0.70 \
  --cool-down-period=90

# Autoscale based on load balancing utilization
gcloud compute instance-groups managed set-autoscaling my-mig \
  --zone=us-central1-a \
  --min-num-replicas=2 \
  --max-num-replicas=10 \
  --target-load-balancing-utilization=0.80

# Autoscale based on Cloud Monitoring metric
gcloud compute instance-groups managed set-autoscaling my-mig \
  --zone=us-central1-a \
  --min-num-replicas=2 \
  --max-num-replicas=10 \
  --custom-metric-utilization=metric=custom.googleapis.com/my_metric,utilization-target=10

# Remove autoscaling
gcloud compute instance-groups managed stop-autoscaling my-mig \
  --zone=us-central1-a
```

Autoscaling Signals

Signal	Description	Use Case
CPU utilization	Average CPU across instances	General workloads
HTTP LB utilization	Requests per second per instance	Web applications
Cloud Monitoring metric	Custom or built-in metrics	Application-specific scaling
Schedule	Time-based scaling	Predictable traffic patterns

Autohealing

Health checks detect unhealthy VMs and automatically recreate them:

```
# Create a health check
gcloud compute health-checks create http my-health-check \
  --port=80 \
  --request-path=/health \
  --check-interval=30s \
  --timeout=10s \
  --healthy-threshold=2 \
  --unhealthy-threshold=3

# Apply health check to MIG for autohealing
gcloud compute instance-groups managed update my-mig \
  --zone=us-central1-a \
  --health-check=my-health-check \
  --initial-delay=300
```

Rolling Updates

Update VMs without downtime:

```
# Update a MIG to use a new template
gcloud compute instance-groups managed rolling-action start-update my-mig \
  --version=template=my-new-template \
  --zone=us-central1-a \
  --max-surge=3 \
  --max-unavailable=0

# Canary update (partial rollout)
gcloud compute instance-groups managed rolling-action start-update my-mig \
  --version=template=my-template \
  --canary-version=template=my-new-template,target-size=20% \
  --zone=us-central1-a
```

MIG Types

Type	Scope	Use Case
Zonal MIG	Single zone	Simple deployments
Regional MIG	Multiple zones in a region	High availability

6. Configuring OS Login

What Is OS Login?

OS Login links a user’s Linux account to their Google identity, simplifying SSH access management.

How It Works

- 1. Enable OS Login at the project or instance level
- 2. Grant appropriate IAM role to users
- 3. Users connect with `gcloud compute ssh` — their Google identity is used

IAM Roles for OS Login

Role	Permission
<code>roles/compute.osLogin</code>	Standard SSH access (non-root)
<code>roles/compute.osAdminLogin</code>	SSH access with sudo (root) privileges

Setup

```
# Enable OS Login project-wide
gcloud compute project-info add-metadata \
  --metadata enable-oslogin=TRUE

# Enable OS Login with 2FA
gcloud compute project-info add-metadata \
  --metadata enable-oslogin=TRUE,enable-oslogin-2fa=TRUE

# Grant OS Login role
gcloud projects add-iam-policy-binding PROJECT_ID \
  --member="user:alice@example.com" \
  --role="roles/compute.osLogin"
```

OS Login vs. Metadata SSH Keys

Feature	OS Login	Metadata SSH Keys
Identity management	IAM-based	Manual key management
2FA support	Yes	No
Audit trail	IAM audit logs	Limited
Granularity	Per-user IAM roles	Project or instance level
Recommended	Yes (Google best practice)	Legacy method

7. Configuring VM Manager

What Is VM Manager?

A suite of tools for managing operating systems on large VM fleets:

Components

OS Patch Management

- Automate OS patch deployment across VM fleets
- Schedule patch windows
- Patch compliance reporting

```
# Create a patch job
gcloud compute os-config patch-jobs execute \
  --instance-filter-names="zones/us-centrall-a/instances/vm-1,zones/us-centrall-a/instances/vm-2" \
  --duration=3600s

# Create a recurring patch deployment
gcloud compute os-config patch-deployments create my-patch-deployment \
  --instance-filter-all \
  --recurring-schedule-frequency=WEEKLY \
  --recurring-schedule-day-of-week=SUNDAY \
  --recurring-schedule-time-of-day='02:00' \
  --recurring-schedule-time-zone='America/New_York'
```

OS Inventory Management

- View OS information, installed packages, and system configurations
- Query across your VM fleet

```
# List OS inventory for instances
gcloud compute os-config inventories list \
  --location=us-centrall-a

# Describe OS inventory for a specific instance
gcloud compute os-config inventories describe my-vm \
  --location=us-centrall-a
```

OS Config Agent

- Must be installed on VMs for VM Manager to work
- Pre-installed on many Google-provided images
- Communicates with VM Manager API

Enabling VM Manager

```
# Enable the OS Config API
gcloud services enable osconfig.googleapis.com

# Set metadata to enable the OS Config agent
gcloud compute project-info add-metadata \
  --metadata=enable-os-config=TRUE

# Verify agent is running on a VM
sudo systemctl status google-osconfig-agent
```

8. Preemptible and Spot VMs

Preemptible VMs (Legacy)

- Up to **91% cheaper** than standard VMs
- Maximum runtime of **24 hours** — Google terminates them after 24 hours regardless
- May be **preempted at any time** with a 30-second shutdown notice
- **Cannot live-migrate** — must use `--maintenance-policy=TERMINATE`

Spot VMs (Current Recommendation)

- Successor to preemptible VMs
- **No 24-hour runtime limit** — but can still be preempted at any time
- Same low pricing as preemptible
- Same restrictions: no live migration, 30-second shutdown notice

Use Cases

- Batch processing jobs
- Fault-tolerant and stateless workloads
- CI/CD pipeline workers
- HPC and data analytics

gcloud Flags

```
gcloud compute instances create my-spot-vm --machine-type=n2-standard-4 --provisioning-model=SPOT
```

- `--provisioning-model=SPOT` — creates a Spot VM (recommended)
- `--preemptible` — legacy flag; creates a preemptible VM
- `--instance-termination-action` — `STOP` (default for Spot) or `DELETE`

Preemption Handling

- On preemption, GCP sends a **shutdown signal** → VM has ~30 seconds to run a shutdown script
- Check instance metadata to detect preemption: `instance/preempted` returns `TRUE` after a preemption event
- Use **Pub/Sub** or **Cloud Monitoring** to trigger alerts or replacement logic on preemption events

9. Snapshots and Custom Images

Snapshots

- **Point-in-time incremental backup** of a persistent disk
- First snapshot is a full backup; subsequent snapshots are incremental (only changed blocks)
- Stored in **Cloud Storage** (billed at Cloud Storage rates, not as a persistent disk)

```
gcloud compute disks snapshot DISK_NAME --zone=ZONE --snapshot-names=SNAPSHOT_NAME --storage-class=STANDARD
gcloud compute snapshots list
gcloud compute snapshots describe SNAPSHOT_NAME
gcloud compute disks create NEW_DISK --source-snapshot=SNAPSHOT_NAME --zone=ZONE --size=10GB
gcloud compute snapshots delete SNAPSHOT_NAME
```

Cross-reference: Snapshot schedules (automated recurring snapshots) are covered in file 12.

Custom Images

- Create from a **stopped VM's disk**, an existing **snapshot**, or a **Cloud Storage file**
- Images belong to a project but can be shared across projects via IAM
- **Image families:** a logical group of images; `--image-family` always resolves to the latest non-deprecated image in that family

```
# Create image from stopped VM disk
gcloud compute images create my-custom-image --source-disk=MY_DISK --source-disk-zone=ZONE
# Create from snapshot
gcloud compute images create my-image --source-snapshot=SNAPSHOT_NAME
# Deprecate old image
gcloud compute images deprecate OLD_IMAGE --state=DEPRECATED --replacement=NEW_IMAGE
# List images in a family
gcloud compute images list --filter="family=my-app-v2"
# Share image with another project
gcloud compute images add-iam-policy-binding my-image --member="serviceAccount:SA@OTHER_PROJECT" --role=roles/imageViewer
```

10. Shielded VMs

Shielded VMs protect against rootkits, bootkits, and firmware-level attacks by verifying the integrity of the VM's boot sequence.

Three Security Features

Feature	What It Does
Secure Boot	Only signed OS kernels and drivers are allowed to load
vTPM (Virtual Trusted Platform Module)	Measures and records boot integrity; enables attestation
Integrity Monitoring	Compares each boot measurement against a known-good baseline; alerts on deviations

Enable at Creation

```
gcloud compute instances create my-vm --shielded-secure-boot --shielded-vtpm --shielded-in
```

Key Restrictions

- Not all machine types or images support Shielded VM — use Shielded-compatible images (e.g., `debian-cloud/debian-12` , `ubuntu-os-cloud/ubuntu-2204-lts`)
- Check image shielded support: `gcloud compute images list --filter="shieldedInstanceInitialState:*`

Exam tip: Shielded VM is a **prerequisite** for **Confidential Computing** (Confidential VMs). If a question asks about hardware-level memory encryption, the answer involves Confidential VMs, which require Shielded VM features.

8. Startup and Shutdown Scripts

Startup Scripts

Run automatically when a VM boots:

```
# Inline startup script
gcloud compute instances create my-vm \
  --metadata=startup-script='#!/bin/bash
  apt-get update
  apt-get install -y nginx
  systemctl start nginx'

# Startup script from Cloud Storage
gcloud compute instances create my-vm \
  --metadata=startup-script-url=gs://my-bucket/startup.sh

# Startup script from local file
gcloud compute instances create my-vm \
  --metadata-from-file=startup-script=./startup.sh
```

Shutdown Scripts

Run when a VM is being stopped or deleted (best-effort, ~90 second window):

```
gcloud compute instances create my-vm \
  --metadata=shutdown-script='#!/bin/bash
  echo "Shutting down gracefully..."
  # Drain connections, save state, etc.'
```

Exam Practice Questions

1. You need to deploy a fleet of identical web servers that automatically replace unhealthy instances. What should you create?
 - Answer: Create an **instance template** defining the VM configuration, then create a **managed instance group** with a **health check** for autohealing.
2. Your MIG needs to handle variable traffic, scaling between 2 and 20 instances based on CPU usage. How do you configure this?
 - Answer: Use `gcloud compute instance-groups managed set-autoscaling` with `--min-num-replicas=2`, `--max-num-replicas=20`, and `--target-cpu-utilization=0.70`.
3. A VM has a GPU attached. What must the maintenance policy be set to?
 - Answer: `TERMINATE` — VMs with GPUs cannot be live-migrated, so the maintenance policy must be set to TERMINATE.
4. You want to manage SSH access using corporate Google identities with sudo access and 2FA. How?

- Answer: Enable **OS Login** with 2FA (`enable-oslogin=TRUE` , `enable-oslogin-2fa=TRUE`), grant `roles/compute.osAdminLogin` for sudo access.

5. You need to update a MIG's template without any downtime. How should you proceed?

- Answer: Create a new instance template, then use `rolling-action start-update` with `--max-unavailable=0` and `--max-surge=3` for a zero-downtime rolling update.

6. How can you connect via SSH to a VM that has no external IP address?

- Answer: Use `gcloud compute ssh my-vm --tunnel-through-iap`. This tunnels the SSH connection through **Identity-Aware Proxy**. Ensure the IAP firewall rule allows TCP from `35.235.240.0/20`.

Glossary

Attestation — The process by which a Shielded VM cryptographically proves the integrity of its boot sequence to a verifier using measurements recorded by the vTPM.

Auto-Delete — A Compute Engine disk setting controlling whether a persistent disk is automatically deleted when its attached VM is deleted. Boot disks default to auto-delete on; additional disks default to off.

Autohealing — A Managed Instance Group feature that automatically detects unhealthy VM instances via health checks and recreates them to restore the group to its desired state.

Autoscaling — A MIG feature that automatically increases or decreases the number of VM instances in the group based on configurable signals such as CPU utilization, HTTP LB utilization, Cloud Monitoring metrics, or a schedule.

Availability Policy — The configuration on a Compute Engine VM that determines behavior during host maintenance events: either live-migrate the VM (`MIGRATE`) or terminate it (`TERMINATE`).

BYOL (Bring Your Own License) — A licensing model allowing customers to use their own software licenses on cloud infrastructure, often requiring sole-tenant nodes for compliance.

Canary Update — A rolling update strategy in which a new template version is deployed to a small percentage of instances first, allowing validation before full rollout.

CI/CD (Continuous Integration / Continuous Delivery) — A software development practice that automates code integration, testing, and deployment. Spot VMs are commonly used as cost-effective CI/CD workers.

Cloud Monitoring — GCP's managed service for collecting, storing, and visualizing metrics and logs from GCP services and custom applications. Supports custom autoscaling metrics for MIGs.

Cloud Storage — GCP's globally unified, scalable object storage service. Snapshots and custom images are stored in Cloud Storage buckets and billed at Cloud Storage rates.

Compute Engine — GCP's Infrastructure as a Service (IaaS) offering that provides virtual machine (VM) instances running on Google's infrastructure.

Confidential Computing (Confidential VM) — A Compute Engine feature that uses hardware-level memory encryption (AMD SEV) to protect data in use. Requires Shielded VM features to be enabled.

Custom Image — A VM boot disk image created by a user from a stopped VM's disk, a snapshot, or a Cloud Storage file. Can be shared across GCP projects via IAM.

Custom Machine Type — A Compute Engine machine type in which the user specifies an arbitrary number of vCPUs and amount of memory (e.g., `custom-6-20480`), rather than using a predefined configuration.

e2 (Machine Family) — A cost-optimized Compute Engine machine family suitable for general-purpose workloads. Includes `e2-micro`, `e2-standard-4`, etc.

ext4 — A widely used Linux filesystem type. Used as an example when formatting newly attached persistent disks inside a VM.

fstab — A Linux configuration file (`/etc/fstab`) that defines how disk partitions and other storage devices are automatically mounted at boot time.

GCP (Google Cloud Platform) — Google's suite of cloud computing services.

gcloud — The primary command-line tool for interacting with GCP services, part of the Google Cloud SDK.

GPU (Graphics Processing Unit) — A specialized processor used for parallel computation, machine learning, and graphics. VMs with GPUs must use `--maintenance-policy=TERMINATE` because they cannot be live-migrated.

Health Check — A probe that periodically tests the responsiveness of a VM or service endpoint. Used by MIGs for autohealing and by load balancers for traffic routing.

HPC (High-Performance Computing) — Computational workloads requiring large amounts of processing power, often parallelized. Spot VMs are cost-effective for HPC batch jobs.

IAM (Identity and Access Management) — GCP's system for controlling who can do what on which resources. OS Login uses IAM roles to control SSH access to VMs.

IAP (Identity-Aware Proxy) — A GCP service that enables secure access to GCP resources without exposing them to the public internet. Used to tunnel SSH connections to VMs that have no external IP.

Image Family — A logical grouping of Compute Engine images; the `--image-family` flag always resolves to the latest non-deprecated image within that family.

Instance Template — A read-only, immutable resource that defines the configuration (machine type, disk, image, network, metadata, service account) used to create VM instances in a Managed Instance Group.

Integrity Monitoring — A Shielded VM feature that compares each boot's integrity measurements against a known-good baseline and alerts on deviations, detecting tampering.

IOPS (Input/Output Operations Per Second) — A measure of storage device performance. Higher IOPS persistent disk types (`pd-ssd` , `pd-extreme`) are used for databases and random I/O workloads.

Label — A key-value metadata pair attached to GCP resources for organization, cost attribution, and filtering (e.g., `environment=production`).

Live Migration — A Compute Engine feature that moves a running VM to a different physical host during maintenance events with no VM downtime. Not supported for VMs with GPUs or Local SSDs.

Local SSD — A physically attached NVMe SSD on the Compute Engine host that provides the highest storage performance. Data is lost when the VM is stopped or terminated.

Machine Type — A predefined or custom configuration specifying the number of vCPUs and amount of RAM for a Compute Engine VM (e.g., `e2-standard-4` , `n2-standard-8`).

Managed Instance Group (MIG) — A group of identical Compute Engine VM instances created from an instance template, managed as a single entity with support for autoscaling, autohealing, and rolling updates.

Metadata — Key-value pairs associated with a GCP project or VM instance. Used to pass configuration data (e.g., startup scripts) to VMs at boot time.

n1 / n2 (Machine Family) — General-purpose Compute Engine machine families. n1 is the first generation; n2 uses Intel Cascade Lake processors with better performance. Both support GPUs.

nginx — A high-performance open-source web server and reverse proxy. Used in examples as a package installed by startup scripts on Compute Engine VMs.

OS Config Agent — A software agent that must be installed on VMs to enable VM Manager features (patch management, inventory). Pre-installed on many Google-provided images.

OS Login — A Compute Engine feature that links a user's Linux account to their Google identity for SSH access, managed via IAM roles. Supports 2FA and provides audit logs.

Patch Deployment — A VM Manager configuration that defines a recurring schedule for applying OS patches to a fleet of VMs (e.g., weekly on Sundays at 02:00).

pd-balanced — A Compute Engine persistent disk type offering a balance of performance and cost; the default disk type for most workloads.

pd-extreme — The highest-performance Compute Engine persistent disk type, supporting up to 120,000 IOPS. Used for mission-critical databases.

pd-ssd — A Compute Engine persistent disk type backed by solid-state drives, offering high IOPS (30 R/W per GB) for databases and latency-sensitive workloads.

pd-standard — A Compute Engine persistent disk type backed by hard disk drives, offering the lowest cost for sequential workloads like logs and bulk storage.

Persistent Disk — A network-attached block storage device for Compute Engine VMs. Data persists independently of the VM lifecycle (unless auto-delete is enabled).

Preemptible VM — A legacy Compute Engine pricing model offering VMs at up to 91% discount. VMs can be preempted at any time with a 30-second notice and have a maximum runtime of 24 hours. Superseded by Spot VMs.

Pub/Sub — GCP's managed, scalable messaging service. Used to receive notifications about Spot VM preemption events.

Read-Only Mode (Persistent Disk) — A mode in which a persistent disk can be attached to multiple VMs simultaneously for shared read access. Write access requires attaching to only one VM.

Regional MIG — A Managed Instance Group that distributes VM instances across multiple zones within a single region, providing higher availability than a zonal MIG.

Rolling Update — A MIG update strategy that replaces instances incrementally using a new instance template, enabling zero-downtime deployments. Controlled by `--max-surge` and `--max-unavailable` parameters.

Secure Boot — A Shielded VM feature that ensures only digitally signed OS kernels and boot components are loaded, protecting against bootkits and rootkits.

Service Account — A special GCP identity used by applications (rather than humans) to authenticate API calls. Assigned to VM instances to grant them GCP API access.

Shielded VM — A Compute Engine security feature that uses Secure Boot, vTPM, and Integrity Monitoring to protect VMs against firmware-level attacks, rootkits, and bootkits.

Snapshot — A point-in-time, incremental backup of a Compute Engine persistent disk stored in Cloud Storage. The first snapshot is a full backup; subsequent ones capture only changed blocks.

Sole-Tenant Node — A dedicated physical server in GCP that runs only your VMs, providing physical isolation for licensing compliance (BYOL), security, or performance requirements.

Spot VM — The current Compute Engine pricing model for fault-tolerant workloads, offering the same deep discounts as preemptible VMs but with no 24-hour runtime limit. Can be preempted at any time with 30 seconds notice.

SSH (Secure Shell) — A cryptographic network protocol for secure remote login to and command execution on VMs. GCP provides several methods for SSH key management.

Startup Script — A script (inline, from a file, or from Cloud Storage) that runs automatically when a Compute Engine VM boots. Specified via instance metadata.

Subnet — A regional IP address range within a VPC network. VMs are created within subnets and receive internal IP addresses from the subnet's range.

VPC (Virtual Private Cloud) — GCP's global, software-defined private network providing networking for VM instances, including subnets, firewall rules, and routes.

vTPM (Virtual Trusted Platform Module) — A virtualized hardware security module used in Shielded VMs to measure and record boot integrity, enabling attestation.

VM Manager — A suite of GCP tools for managing operating systems on large fleets of Compute Engine VMs, including OS Patch Management and OS Inventory Management.

Zonal MIG — A Managed Instance Group in which all VM instances reside in a single zone. Simpler than a regional MIG but susceptible to zone-level failures.